

Week 4

Q1. Write a C++ program to check whether a number is even or odd using ternary operator.

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter number: " << endl;
    cin >> num;
    string result = ((num % 2) == 0) ? "Even":"Odd";
    cout << "The given number is " << result << endl;
    return 0;
}
```

```
Enter number:
100
The given number is Even

-----
Process exited after 4.883 seconds with return value 0
Press any key to continue . . . |
```

Q2. Write a C++ program to perform the addition of two numbers without using + operator.

```
#include <iostream>
using namespace std;

int main () {
    int a, b;
    cout << "Enter values to add: ";
    cin >> a >> b;

    while (b > 0) {
        int c = a & b;
        a = a^b;
        b = c<<1;
    }

    cout << "Sum = " << a;
    return 0;
}
```

```
Enter values to add: 5 6
Sum = 11

-----
Process exited after 24.85 seconds with return value 0
Press any key to continue . . . |
```

Q3. Write a C++ program to evaluate the arithmetic expression $((a + b / c * d - e) * (f - g))$. Read the values a, b, c, d, e, f, g from the standard input device.

```

#include <iostream>
using namespace std;

int main() {
    int a, b, c, d, e, f, g;
    cout << "Enter value of a: " << endl;
    cin >> a;
    cout << "Enter value of b: " << endl;
    cin >> b;
    cout << "Enter value of c: " << endl;
    cin >> c;
    cout << "Enter value of d: " << endl;
    cin >> d;
    cout << "Enter value of e: " << endl;
    cin >> e;
    cout << "Enter value of f: " << endl;
    cin >> f;
    cout << "Enter value of g: " << endl;
    cin >> g;
    float exp = (a+b/c*d-e)*(f-g);
    cout << "Value of given expression: " << exp << endl;
    return 0;
}

```

```

Enter value of a:
5
Enter value of b:
6
Enter value of c:
2
Enter value of d:
1
Enter value of e:
9
Enter value of f:
3
Enter value of g:
8
Value of given expression: 5

-----
Process exited after 47.89 seconds with return value 0
Press any key to continue . . .

```

Q4. A Fibonacci sequence is defined as follows: The first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C++ program to generate the first n terms of the sequence.

```

#include <iostream>
using namespace std;

int main() {
    int n, i, fib = 0;
    cout << "Enter Number: ";
    cin >> n;
    int a=0, b=1;
    for (i=0; i<n; i++) {
        cout << a << " ";
        int temp = a + b;
        a = b;
        b = temp;
    }
    return 0;
}

```

```

Enter Number: 10
0 1 1 2 3 5 8 13 21 34

-----
Process exited after 4.826 seconds with return value 0
Press any key to continue . . . |

```

Q5. Write a C++ program to generate all the prime numbers between 1 and n, where n is a value supplied by the user.

```
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int n, i, j;
    cout << "Enter number: ";
    cin >> n;
    for (i=2; i<=n; i++) {
        int flag = 1;
        for (j=2; j<=sqrt(i); j++) {
            if (i%j==0) {
                flag = 0;
                break;
            }
        }
        if (flag) {
            cout << i << " ";
        }
    }
    return 0;
}
```

```
Enter number: 20
2 3 5 7 11 13 17 19
-----
Process exited after 14.92 seconds with return value 0
Press any key to continue . . .
```

Q6. A character is entered through keyboard. Write a C++ program to determine whether the character entered is a capital letter, a small case letter, a digit or a special symbol using if-else and switch case. The following table shows the range of ASCII values for various characters. Characters ASCII values A– Z: 65– 90, a– z: 97– 122, 0– 9: 48– 57 Special symbols 0– 47, 58– 64, 91– 96, 123– 127

```
#include <iostream>
using namespace std;

int main() {
    char c;
    cout << "Enter Character: ";
    cin >> c;
    if (c > 65 && c < 90) cout << "Character is a Capital Letter";
    else if (c > 97 && c < 122) cout << "Character is a Small Letter";
    else if (c > 48 && c < 57) cout << "Character is Digit";
    else cout << "Character is a Special Character";
}
```

```
Enter Character: ~
Character is a Special Character
-----
Process exited after 6.638 seconds with return value 0
Press any key to continue . . . |
```

Q7. Write a C++ program to find the roots of a quadratic equation.

```
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    double a, b, c;

    cout << "Enter values of a, b, c: ";
    cin >> a >> b >> c;

    if (a == 0) {
        cout << "Not a quadratic equation (a cannot be 0)." << endl;
        return 0;
    }

    double discriminant = b * b - 4 * a * c;

    if (discriminant > 0) {
        double root1 = (-b + sqrt(discriminant)) / (2 * a);
        double root2 = (-b - sqrt(discriminant)) / (2 * a);

        cout << "Two real and distinct roots:\n";
        cout << "x1 = " << root1 << endl;
        cout << "x2 = " << root2 << endl;
    }
    else if (discriminant == 0) {
        double root = -b / (2 * a);

        cout << "Two equal real roots:\n";
        cout << "x1 = x2 = " << root << endl;
    }
    else {
        double realPart = -b / (2 * a);
        double imaginaryPart = sqrt(-discriminant) / (2 * a);

        cout << "Complex roots:\n";
        cout << "x1 = " << realPart << " + " << imaginaryPart << "i" << endl;
        cout << "x2 = " << realPart << " - " << imaginaryPart << "i" << endl;
    }

    return 0;
}
```

```
Enter values of a, b, c: 1 2 5
Complex roots:
x1 = -1 + 2i
x2 = -1 - 2i

-----
Process exited after 7.224 seconds with return value 0
Press any key to continue . . . |
```

Q8. Write a C++ program to check whether a given 3-digit number is an Armstrong number.

```
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int n, sum=0;
    cout << "Enter number: ";
    cin >> n;
    int temp = n;
    while(temp > 0) {
        sum += pow((temp%10),3);
        temp/=10;
    }
    if (sum == n) {
        cout << "Given number is Armstrong number";
    } else {
        cout << "Given number is NOT a Armstrong number";
    }
    return 0;
}
```

```
Enter number: 153
Given number is Armstrong number
-----
Process exited after 4.392 seconds with return value 0
Press any key to continue . . .
```